

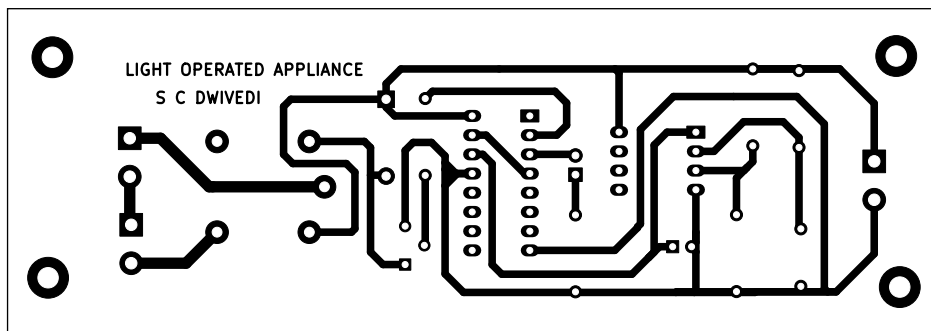


Fig. 3: Actual-size PCB

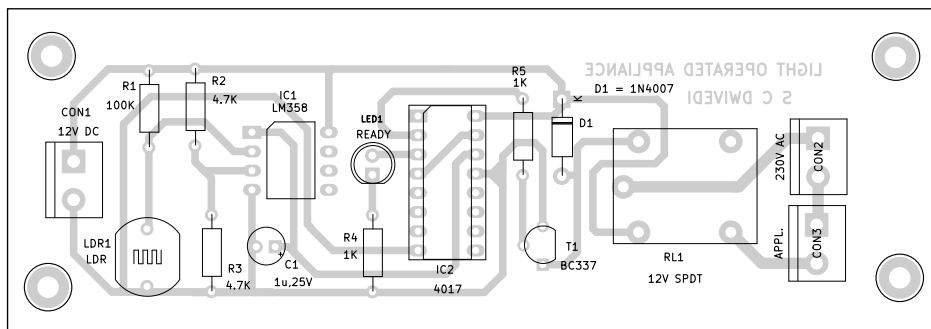


Fig. 4: Components layout

current flow. In bright light or when sufficient light is present, the LDR's resistance drops significantly, allowing more current to flow.

The voltage at pin 3 is fixed using a voltage divider built with resistors R2 and R3, serving as the reference voltage for op-amp IC1A. The LDR senses the surrounding light, and its resistance changes alter the voltage at pin 2 of op-amp IC1A, generating clock pulses for IC2.

The op-amp IC1A is a comparator, generating clock pulses to drive IC 4017. The clock inhibits pin (pin 13) is grounded, enabling IC 4017 to operate freely, while the reset

pin connects to output Q2 (pin 4) for toggling between Q0 and Q1 outputs.

Working is simple. If on powering on the circuit LED1 glows, it means circuit is ready to use and appliance is off. When light falls on the LDR first time, its output provides a clock to CD4017. It advances a clock from Q0 to Q1 and transistor T1 conducts and energises relay RL1 to switch on the appliance. When light falls on the LDR second time, its output provides another clock to CD4017 and again advances a clock from Q1 to Q2, transistor T1 stops conducting and de-energises relay RL1 to

switch off the appliance. At the same time, it resets the circuit and LED1 glows. After that the cycle repeats. A freewheeling diode (D1) safeguards the transistor from the back EMF.

Construction and testing

An actual-size, single-sided PCB layout for the circuit is shown in Fig. 3, with its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box.

Alternatively, the circuit can be assembled on a breadboard. It can then be mounted in a suitable enclosure with openings for the LDR and appliance. A PVC electrical box is recommended. The LDR must be shielded from normal light, allowing only focused light from a mobile or laser

torch to activate it.

To test the system, focus the light on the LDR using a mobile or laser torch. The relay should energise and activate the appliance. When focused again, the relay should de-energise and deactivate the appliance. The circuit can be powered with a 12V DC adaptor or battery. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-light-operated-appliance>

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou

The latest from the Open Source world is here.

OpenSourceForU.com

Join the community at facebook.com/opensourceforu

Follow us on Twitter @OpenSourceForU